

Generation Interconnection Request System Impact Study Report

For Queue Project Q23-2355

Corona Ave 100 kV
February 18, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2355. The Interconnection Customer proposes a 62.6 MW solar photovoltaic facility interconnecting at the Corona Ave 100 kV substation in the Illinois area. The request was received on October 19, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 492 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$285,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Shell Chemical to W Kingston 100.0 kV Circuit 1
Contingency: W Kingston to Shell Chemical 100.0 kV #2
Loading: 105.8% (Rating: 144 MVA)
DFAX: 6.6%
Pre-project loading: 103.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shell Chemical to W Kingston 100.0 kV Circuit 1	Reconductor line	\$585,000	49% (\$285,000)
	Total allocated to project		\$285,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.